Course	**CS/CE 446/452 – Applied Digital Image Processing (L1)**
Semester	Fall 2026
Instructor	Dr Imran Naseem
Document	Project Proposal (Proposal Performa)
Institution	Habib University

1. Project Title

Aviation-Oriented Crack Segmentation: A Multi-Method Classical Image Processing Pipeline with Rivet- and Seam-Aware False-Positive Suppression

Domain: aerospace structural inspection and civil structural health monitoring. Approach: classical digital image processing only (no deep learning).

2. Team Members

- **Emaan Asad**
- **Lamia Munir Shaikh**
- **Ebad Atif**

3. Problem Statement

Detecting surface cracks on engineered structures is still largely a manual, visual task. In civil structural health monitoring this makes inspection slow and subjective; in aviation maintenance it is a safety-critical bottleneck, because aircraft skin must be walked and examined panel by panel during line and base checks. Fatigue cracks in aluminium skin characteristically initiate at stress concentrations around rivets and fastener holes, which is precisely where automated detection is hardest: rivet heads, panel seams, lap-joint edges, paint boundaries and scratches are all strong intensity discontinuities, so a conventional edge-based detector reports them as cracks. Published classical crack-detection work on aircraft structures documents exactly this failure, with rivet holes and specimen-hole edges mistakenly marked as cracks.

This project therefore develops an interpretable, lightweight, fully classical DIP pipeline that (i) segments cracks at the pixel level, (ii) compares structurally different classical segmentation families rather than relying on a single thresholding step, and (iii) adds a geometry-driven suppression stage that explicitly removes fastener- and seam-induced false positives. Deep learning is deliberately excluded: the objective is a transparent, parameter-inspectable algorithm whose every decision can be traced to a measurable image property, which also suits the safety-certification culture of aviation inspection and the available pixel-labelled data volume.

4. Objectives

1. **Preprocessing benchmark.** Quantitatively evaluate denoising and contrast-enhancement front ends using PSNR and SSIM, and select the combination that maximises crack-to-background contrast without blurring crack edges.

   Anisotropic diffusion + CLAHE, evaluated against the raw grayscale baseline.

2. **Multi-method segmentation benchmark.** Implement and compare structurally distinct classical segmentation approaches on an identical image set and identical ground truth, reporting Dice, IoU, Precision, Recall and F1, with a target best-method F1 >= 0.75 on concrete imagery.

   Two methods Canny + adaptive threshold, and Frangi ridge enhancement which already satisfy the course requirement to go beyond a single method.

3. **Structural false-positive suppression for aircraft skin.** Design and evaluate a Hough- and shape-based stage that suppresses rivets, fasteners, panel seams and scratches, targeting >= 40% reduction in false-positive pixels on aircraft-skin images for <= 5% loss of crack recall.

   Circular-Hough fastener localisation and line-Hough seam suppression, with/without ablation.

4. **Quantitative measurement and cross-domain robustness.** Extract calibrated crack geometry from the skeleton, and quantify how the pipeline degrades across surface types and under controlled perturbations.

   Calibrated length and max/mean width; degradation measured on concrete vs. aircraft skin under one or two perturbations (e.g. shadow, blur).

5. Proposed Methodology

The project proceeds through five stages. Stages 1, 4 and 5 form a conventional, reproducible crack-measurement pipeline. Stages 2 and 3 carry the non-trivial contribution: a controlled comparison of multiple classical segmentation families, and an aviation-specific suppression stage. Every item below is tagged Required (the graded core, sized to fit comfortably in 14 weeks) or Optional (genuine additions that strengthen the project but can be cut, reduced, or deferred without breaking anything else if a stage runs long). The table summarises the same split before the stage-by-stage detail.

Component	Required (core, graded baseline)	Optional / reducible if time is short
Stage 1 — Preprocessing	Grayscale conversion; anisotropic diffusion; CLAHE.	Gaussian/median comparison baselines (drop, keep anisotropic diffusion only); morphological top-hat (aviation-only — drop if the aircraft-skin transfer itself is dropped).
Stage 2 — Segmentation methods	(a) Canny + adaptive threshold and (b) Frangi ridge enhancement — two structurally different methods, satisfying the course's 'beyond a single method' requirement.	(c) DWT multi-resolution and (d) texture+k-means — implement whichever fits the remaining schedule; doing only one of the two is an acceptable reduction, doing neither still leaves two methods compared.
Stage 3 — Aircraft-specific suppression	Fastener localisation (Circular Hough) and panel-seam suppression (Line Hough + morphological opening) — the	Shape-based component discrimination (eccentricity/solidity/tortuosity rule filter) — reduce to a simple area+aspect-ratio

Component	Required (core, graded baseline)	Optional / reducible if time is short
	minimum needed to justify the aviation framing at all.	filter; structure-line guided continuity refinement — drop entirely, it is the most complex and most removable single item in the whole proposal.
Stage 4 — Post-processing	Morphological closing, small-object removal, skeletonization.	Spur pruning — keep if time allows, otherwise accept slightly inflated length measurements and note it as a limitation.
Stage 5 — Measurement & robustness	Calibrated crack length and width from the skeleton.	Orientation/tortuosity histogram — nice-to-have, not required for the metrics in Objective 4; full robustness grid (3 surfaces x 5 perturbations) — reduce to 1 surface pair (concrete vs. aircraft) x 2 perturbations (shadow, blur) if weeks 12–13 are tight.
Domain scope	Concrete + aircraft skin (two use cases, as scoped in Section 3).	Pavement/bridge as a third surface (Section 7 dataset) — drop first if anything must go; it only supported the optional cross-domain grid above.
Deliverable format	Modular Python script, per-stage, runnable from the command line.	Lightweight GUI (PyQt/Streamlit) — cut without any loss to the graded methodology.

How to use this. Build Required items first, in stage order — each later Required item depends only on earlier Required output, never on an Optional one, so nothing breaks if an Optional item is skipped. Add Optional items only once the Required chain for that stage already runs end-to-end. If week 9 arrives and the team is behind, cut from the bottom of the Optional column up, not stage-by-stage — e.g. drop the pavement surface and the GUI before touching Stage 3's core Hough detection.

Stage 1 — Preprocessing and Enhancement

Goal: suppress irrelevant surface texture and normalise illumination while preserving crack edges.

1. *Grayscale conversion.* Cracks are intensity features, not chromatic ones; a single channel also removes paint-colour variation across aircraft panels.
2. *Noise suppression — anisotropic diffusion (Perona–Malik).* Chosen over Gaussian blur because it smooths within homogeneous regions while inhibiting diffusion across high-gradient boundaries, so surface grain and rivet-sealant speckle are reduced without softening the thin, low-contrast crack edges the later stages depend on.
3. *Contrast enhancement — CLAHE.* Contrast Limited Adaptive Histogram Equalization boosts local contrast of faint hairline cracks; the clip limit prevents the noise amplification that plain histogram equalization would cause on smooth metal panels.

Stage 2 — Multi-Method Classical Segmentation Benchmark (core comparison)

Goal: isolate crack pixels as a binary mask, and establish which classical technique family actually works best on which surface. All methods consume the identical Stage 1 output and are scored against the identical ground truth, so differences are attributable to the segmentation principle rather than to preprocessing.

(a) Canny edge detection + adaptive thresholding — gradient-based baseline. Canny provides low error rate and thin, well-localised edges; local adaptive thresholding (Otsu or Gaussian-adaptive, computed per region) handles uneven illumination that a single global threshold cannot. This is the reference against which the others are measured.

(b) Frangi / Hessian ridge enhancement — curvilinear-structure-based. The Frangi vesselness filter analyses the eigenvalues of the Hessian at multiple scales to respond specifically to elongated tubular or ridge-like structures. Cracks are exactly such structures, so this method targets crack shape rather than crack edges, and is expected to be more robust to isolated dark blobs (stains, dents, rivet shadows) that gradient methods accept. Together, (a) and (b) already satisfy the course's 'beyond a single method' requirement on their own.

Stage 3 — Rivet- and Seam-Aware False-Positive Suppression (aviation-specific)

Goal: remove the structural and cosmetic features that the segmentation methods misclassify as cracks on aircraft skin. This stage is the project's main original component and is driven purely by geometry and morphology.

4. *Fastener localisation via Circular Hough Transform.* Rivet heads and fastener holes are near-circular with a narrow radius range. Detecting them explicitly serves two purposes: their annuli are excluded from crack scoring, and — following the ROI-minimisation strategy used in radiographic aircraft wing inspection — the region immediately surrounding each fastener becomes a high-priority search window, since that is where fatigue cracks actually initiate.
5. *Panel seam and lap-joint suppression.* Seams are long, straight, high-contrast lines. A Hough Line Transform, combined with morphological opening using long linear structuring elements at multiple orientations, identifies them for removal. Crack paths are tortuous and rarely survive an opening with a long straight element, so this separates the two classes structurally.

An ablation is planned: every metric in Objective 3 is reported with and without Stage 3, so its contribution is isolated and measurable rather than assumed.

Stage 4 — Post-Processing and Refinement

6. *Morphological closing* with a small (e.g. 3x3) structuring element to bridge minor breaks in crack continuity.
7. *Small-object removal* via connected-component area thresholding, deleting isolated noise and residual texture speckle.
8. *Skeletonization* using the Zhang–Suen thinning algorithm to reduce the mask to a single-pixel-wide centreline.

Stage 5 — Quantitative Analysis and Robustness Evaluation

9. *Crack length.* Sum of skeleton pixels along the centreline, converted to millimetres using a pixel-per-mm calibration constant derived from a known reference object or fastener diameter in the frame.
10. *Crack width.* At sampled points along the skeleton, search perpendicular to the local skeleton tangent until the mask boundary is reached, yielding maximum and mean width in millimetres.
11. *Robustness stress test (reduced grid).* Re-evaluate the pipelines actually built (two methods minimum, with and without Stage 3) on the two Required surfaces — concrete and aircraft skin — under one or two perturbation types, e.g. synthetic shadow and defocus blur.

6. Expected Outcomes

Deliverable	Description	Measurement / Metric	Tier
Preprocessing study	Comparison of denoising and contrast-enhancement front ends (anisotropic diffusion + CLAHE required; Gaussian/median comparison and top-hat optional).	PSNR and SSIM against clean references; qualitative crack-visibility panels.	Required core / Optional extras
Multi-method segmentation benchmark	Two classical segmentation families required (Canny+adaptive threshold, Frangi); DWT and texture+k-means optional, extending two methods to up to four.	Dice, IoU, Precision, Recall, F1 per method; target best-method F1 >= 0.75 on concrete.	Required core / Optional extras
Rivet- and seam-aware suppression	Fastener and seam suppression required; shape-based discrimination and structure-line refinement optional refinements.	False-positive pixel reduction (target >= 40%) at recall loss <= 5%; per-class confusion on aircraft images.	Required core / Optional extras
Cross-domain robustness study	Required: concrete + aircraft skin under 1–2 perturbations. Optional: add pavement/bridge as a third surface and extend to the full five-perturbation grid.	IoU/F1 degradation curves per surface type and per perturbation level.	Required core / Optional extras
Quantitative crack geometry	Length and width required; tortuosity/orientation histogram optional.	Total length (mm), maximum and mean width (mm); tortuosity, orientation histogram if time allows.	Required core / Optional extras
Visual output	Segmented mask overlaid on the source image, with suppressed regions colour-coded.	Side-by-side raw / mask / overlay figures for a fixed test subset.	Required
Code implementation	Modular Python package with a per-stage CLI.	Reproducible script: raw image in, mask plus measurement report out.	Required

Scope and risk note. The publicly available aircraft-skin defect datasets are annotated with bounding boxes for five defect classes (crack, dent, scratch, paint-peel, missing-head), not with pixel-level masks. Pixel-level ground truth for the aircraft subset must therefore be traced manually by the team; we budget a subset of roughly 60–80 crack-containing aircraft images for this, and use the larger, already mask-annotated concrete datasets for parameter tuning and for the primary Dice/IoU benchmark. If annotation effort or dataset access proves limiting, Stages 1, 2, 4 and 5 on concrete imagery stand alone as the guaranteed core deliverable, with the aircraft-skin transfer and Stage 3 suppression presented as the extended contribution.

7. Tools & Resources

Language and libraries: Python 3 with OpenCV (Canny, Hough transforms, morphology, adaptive thresholding), scikit-image (Frangi filter, Zhang–Suen skeletonization, GLCM/LBP texture features, region properties), PyWavelets (discrete wavelet decomposition — optional Stage 2c only), NumPy and SciPy (numerics, anisotropic diffusion), scikit-learn (k-means clustering — optional Stage 2d only), Matplotlib (figures), and optionally PyQt or Streamlit for a lightweight demo GUI.

Datasets:

- **Concrete Crack Images for Segmentation (Mendeley Data)** — mask-annotated concrete cracks; primary source for tuning and the pixel-level benchmark.
- **Özgenel / CConCrack concrete crack dataset (Kaggle)** — additional concrete imagery for volume and variety
- **Aircraft skin defect datasets (Roboflow Universe)** — two public collections of aircraft surface defect images labelled into crack, dent, scratch, paint-peel-off and missing-head classes; the smaller was manually photographed across three aircraft types and the larger contains roughly 1,100 images. Used for the aviation-domain transfer and Stage 3 evaluation, with pixel-level masks traced in-house as noted above.

Hardware: standard laptop CPUs. The pipeline is intentionally lightweight; no GPU or training infrastructure is required, which is part of the point.

8. Project Timeline (14 weeks)

Weeks	Activity	Milestone / Output
1–2	Dataset acquisition and curation; pixel-level ground-truth annotation of the aircraft-skin subset; repository and evaluation harness set up.	Curated dataset + annotated masks + working metrics module.
3–4	Stage 1: preprocessing (anisotropic diffusion, CLAHE required; top-hat optional).	Selected preprocessing front end, justified by PSNR/SSIM.

Weeks	Activity	Milestone / Output
5–7	Stage 2: Canny+Frangi required; DWT and/or texture+k-means optional if time allows.	Comparison table across two (required) to four (optional) methods.
8–9	Stage 3: fastener + seam suppression required; shape discrimination and structure-line refinement optional.	False-positive reduction measured on aircraft-skin images.
10	Stage 4: morphological refinement, skeletonization required; spur pruning optional.	Clean single-pixel skeletons for measurement.
11	Stage 5: length/width required; orientation/tortuosity optional.	Length/width report per test image (orientation if time allows).
12–13	Reduced robustness test required (concrete vs. aircraft, 1–2 perturbations); full 3-surface x 5-perturbation grid and pavement dataset optional.	Degradation curves and ablation table.
14	Final report, demo, and presentation.	Report + code + demo + slides.

9. References

[1] X. Wang, B. S. Wong, C. S. Tan, and C. G. Tui, "Automated crack detection for digital radiography aircraft wing inspection," *Research in Nondestructive Evaluation*, vol. 22, no. 2, pp. 105–127, 2011. (Aviation methodology anchor: ROI minimisation plus improved Canny to detect cracks around fastener holes.)

[2] B. S. Wong, X. Wang, C. M. Koh, C. G. Tui, C. Tan, and J. Xu, "Crack detection using image processing techniques for radiographic inspection of aircraft wing spar," *Insight – Non-Destructive Testing and Condition Monitoring*, vol. 53, no. 10, pp. 552–556, 2011. (Documents rivet holes being misdetected as cracks — the failure Stage 3 addresses.)

[3] Y. Zhang et al., "Robust surface crack detection with structure line guidance," *International Journal of Applied Earth Observation and Geoinformation*, vol. 124, art. 103527, 2023. (Structure-line continuity refinement, Stage 3 — optional item.)

[4] A. F. Frangi, W. J. Niessen, K. L. Vincken, and M. A. Viergever, "Multiscale vessel enhancement filtering," in *MICCAI 1998*, LNCS vol. 1496, pp. 130–137. (Hessian ridge filter, Stage 2b — required.)

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[7] P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 12, no. 7, pp. 629–639, 1990. (Stage 1 denoising — required.)

[8] N. Otsu, “A threshold selection method from gray-level histograms,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979. (Binarisation, Stage 2a.)

[9] K. Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, Academic Press, 1994, pp. 474–485. (CLAHE, Stage 1.)

[10] N. Safaei et al., “An automatic image processing algorithm based on crack pixel density for pavement crack detection and classification,” *International Journal of Pavement Research and Technology*, vol. 14, pp. 475–487, 2021. (Classical pavement crack baseline.)

[11] X. Xie et al., “Sparse-sensing and superpixel-based segmentation model for concrete cracks,” *Computer-Aided Civil and Infrastructure Engineering*, 2022. (Published accuracy and robustness bar cited for comparison only; not implemented.)

[12] Aircraft skin defect datasets, Roboflow Universe (open datasets, CC BY 4.0), 2023. (Aviation-domain imagery; crack, dent, scratch, paint-off and missing-head classes.)